BRAD BURCH

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SUMMARY

Senior full-stack software engineer with 7 years building production systems at Salesforce, Total Brain, Front, and Pawservation. Architected and own a booking, payments, and Claude agent platform (MCP, OAuth 2.1 + PKCE, Stripe Connect) on serverless infrastructure that moved more than half of bookings to self-serve, from data model through evals, on-call, and incident response.

SKILLS

Languages & Frameworks: TypeScript, Python, JavaScript, SQL, Java, Scala, React 19, Node.js, Django, Hono, Spring, Play
Infrastructure: PostgreSQL, SQLite (D1), AWS (DynamoDB, SQS), serverless (Cloudflare Workers, Durable Objects, Queues), workflow orchestration (Cloudflare Workflows), R2, MSSQL, Docker, Kubernetes, REST APIs, webhooks, distributed systems
AI / LLM: Claude (Haiku, Sonnet), Vercel AI SDK, Model Context Protocol (MCP), tool calling, LLM evals, prompt caching, prompt-injection defense, AI cost metering, Claude Code, Cursor
Payments & Security: Stripe (Connect, Billing, Checkout), OAuth 2.1 + PKCE, JWT, SAML 2.0, HMAC webhook verification, AES-GCM, SSRF prevention, multi-tenant isolation
Testing & Delivery: Vitest (Node and Workers runtime), Playwright, contract tests, GitHub Actions, Jenkins, CI/CD, alerting

EXPERIENCE

Software Engineer 2022 - Present
Pawservation, San Francisco, CA (pawservation.com) | booking, payments, and AI platform for pet-care businesses

- Architected a production TypeScript booking and payments platform (React, Hono, serverless Cloudflare Workers, SQLite, durable workflow orchestration) serving 30+ active clients and 60+ bookings/month.
- Shipped a Claude Haiku 5.5 booking agent on web chat and WhatsApp plus a remote MCP server from one set of tool definitions, moving more than half of bookings to self-serve.
- Prevented double charges on post-stay card payments by claiming a database uniqueness constraint before calling Stripe, verified with 20 concurrent writers.
- Ended false delinquency flags by matching each payment to its closest booking, so a $40 payment no longer clears a $600 stay and prepayments count.
- Hand-built an OAuth 2.1 + PKCE authorization server for an AI agent (S256, rotating hashed refresh tokens, SSRF-guarded client metadata), letting clients book directly from their own AI agent.
- Cut eval-run cost 79% by restructuring prompts into cached static and dynamic blocks, serving about 95% of input tokens from cache with pass rate unchanged.
- Isolated tenant data across config, pricing, capacity, and bookings by scoping every SQL query through one business-id module, with cross-tenant leak tests on overlapping client records.
- Restored monthly client invoicing after diagnosing four months of truncated runs from CPU analytics (154 ms to 18 ms per PDF), then added completion markers and catch-up windows so missed months self-heal.
- Prevented double-booked stays across four booking channels with an expiring lock on the shared calendar and an overbooking re-check at approval.

Software Engineer 2022
Front, San Francisco, CA (role eliminated in company-wide layoff)

- Shipped a Node.js REST API enabling account-level email rules for multi-domain enterprise customers, previously configurable only per-inbox.
- Shipped a security upgrade to the core JavaScript email library handling 200K+ outbound messages per month.

Software Engineer 2019 - 2021
Total Brain, San Francisco, CA

- Owned enterprise integration delivery as engineering point of contact on Customer Success and Sales calls, standardizing SAML 2.0 and JWT SSO plus HubSpot and Box integrations to cut onboarding from 2+ weeks to ~3 days.
- Migrated user data from MSSQL to DynamoDB with a redesigned NoSQL schema, cutting average API response times by 50%, and mentored junior engineers on backend patterns.
- Shipped a Scala backend integration for Apple, Google, and Facebook authentication, contributing to a 60% lift in new sign-ups.

Software Engineer 2017 - 2018
Salesforce, San Francisco, CA

- Led delivery of a config builder and validator used by internal teams, scoping milestones and coordinating partner teams.
- Built an OAuth2 security filter for microservice access and a React management UI that cut internal test setup time 40%.

PROJECTS

- **MCP Auth Kit (open source):** OAuth 2.1 + PKCE kit for MCP servers with rate limits, scoped tools, and two-phase confirmation.
- **Roadrunner (open source):** Python/Django and PostgreSQL app on Strava OAuth and webhooks, matching eBird and iNaturalist sightings to activities with a 2/4/8-hour deferred re-check queue.

EDUCATION

DePauw University, Bachelor of Arts in Computer Science 2017